

Builder Basics · Build a Village — English Worksheet

Topic: Build a Village (Mid-Course Project) · **Course:** Builder Basics · **Time:** about 60 minutes

1 · Predict 🌟

Read each set of steps. Write what you think will happen.

```
set size to 5
set height to 3

repeat height times:
  repeat 4 times:
    repeat size times:
      place block down
      move forward
    turn right
  move up by 1
```

Which part of the village is this — house, well, or farm house?


```
repeat 4 times:
  repeat 3 times:
    place block down
    move forward
  turn right
repeat 3 times:
  destroy block down
  move down by 1
```

Which part of the village is this?


```
# Stage 1
[house walls + roof]
# Stage 2
move to next spot
[well]
# Stage 3
move to next spot
[farm house]
```

This is the skeleton for the whole village. What does the agent need between each stage?

2 · Spot the Bug 🐛

Each code block is broken. Read what it should do, rewrite it, then explain why the original was wrong.

Bug A — The village should have **a house with a roof, a well, and a farm house** — three structures, side by side.

```
# House
[house walls + roof code]
# Well
[well code]
```

Hint: something is missing. Rewrite the skeleton to include all three structures.

Write the fixed skeleton:

What was missing? Why does the full version work?

Bug B — Between each structure, the agent must **walk to a new spot** so the next structure doesn't land on top of the last one.

```
# House
[house walls + roof]
# Well
[well code]
# Farm house
[farm house code]
```

Hint: add the missing "move to next spot" steps. You don't need to write the full house/well/farm code — write where the agent should walk between stages.

Write the fixed skeleton:

Why was the original wrong? Why does your fix work?

Bug C — The house roof should sit on top of the walls, but in this code the agent stays on the ground after the walls.

```
repeat height times:
    repeat 4 times:
        repeat size times:
            place block down
            move forward
        turn right
# now place the roof
repeat size times:
    repeat size times:
        place block down
        move forward
```

Hint: the agent finishes the walls on the ground. Where should it go before placing the roof?

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world and build your village — **a house with a roof, a well, and a farm house**. If you have time, add a standalone farm too.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.